Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

ML Engineer with 8+ years of production Python engineering in the **property leasing domain** — the exact domain AppFolio builds for. Built **Leasel**: an agentic LangGraph RAG pipeline for lease compliance (93.6% precision, 87.9% F1, pass@k eval harness). Built **AgentProbe**: open-source multi-agent testing toolkit with ChromaDB RAG, 5-metric LLM-as-judge eval engine, and CI/CD quality gates. Shipped Entrata's first production AI pipeline (PyTorch + OpenAI API) across 1.6M financial records and automated affordable housing compliance across 100K+ unit portfolios. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 – Aug 2024

Trainee → Associate Engineer → Software Engineer

- Automated **income eligibility verification, AMI threshold calculations, and recertification workflows** across 100K+ unit portfolios by building Section 8/HCV, LIHTC, and HUD compliance modules — zero compliance violations.
- Shipped Entrata's **first production AI pipeline** across 1.6M real financial records — output validation, quality gates, and compliance controls at launch; later productized into a customer-facing feature (Python, PyTorch, OpenAI API).
- Reduced deployment times **40%** and sustained **99.5% uptime** across 6+ months by shipping 20+ full-stack modules (Python, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users.
- Cut microservices execution time **50%** by leading Docker containerization; improved query performance **30%** across 200+ SQL queries; achieved **100% PHI compliance** on AWS S3 migration — zero security incidents.

George Mason University

Aug 2024 – Aug 2025

Research Associate — Statistical Modeling · ML Research

- Achieved **$R^2 = 0.977$** regression accuracy across 190+ countries by building multi-source prediction models (Python, R) on WHO and UK Road Safety datasets; surfaced anomalies manual review missed across 3 research pipelines by applying Isolation Forest.

PROJECTS

Leasel: Agentic RAG Pipeline for Lease & Affordable-Housing Compliance

Python · LangGraph · LLaMA · Transformers · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Achieved **93.6% precision and 87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs by building an agentic LangGraph RAG pipeline — 5 US states + federal standards, 1,050 synthetic tenants, ~40% injected violations.
- Reached **pass@1 81.7% and pass@3 88.3%** by implementing a pass@k evaluation harness with multi-trial sampling, regression gate, and LLM-as-judge scoring across accuracy and reasoning quality — deterministic rules engine keeps all compliance determinations fully auditable.

AgentProbe: Multi-Agent AI Evaluation & Observability Toolkit

Open Source

Python · LangGraph · LangChain · FastAPI · ChromaDB · LLM-as-Judge · Anthropic · OpenAI · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Blocked failing agent runs before production by building a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD integration — every commit evaluated automatically.
- Delivered trace-level observability across concurrent agent test suites — per-run metrics, regression drift, retrieval quality, and latency — by building a React dashboard backed by a **ChromaDB RAG pipeline**.

CrashClarity: Large-Scale Accident Severity Prediction (104K Records)

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering — Live Dashboard

- Achieved **85%+ accuracy** on 104K+ accident records by building a Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables — deployed interactive dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

ML & AI: PyTorch, Scikit-learn, XGBoost, Random Forest, Logistic Regression, Isolation Forest, LangGraph, LangChain, RAG Pipelines, ChromaDB, FAISS, Vector Databases, LLM-as-Judge, LLM Evaluation, Embeddings, NLP, spaCy, BERT, Transformers, Hugging Face, Feature Engineering, Anomaly Detection

Languages: Python, SQL, PL/SQL, JavaScript, TypeScript, R, Bash

APIs & Backend: FastAPI, REST APIs, OpenAI API, Anthropic API, Groq, Ollama, React, RabbitMQ

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), PostgreSQL, MySQL, SQLite, Git, Linux, Agile/Scrum (120+ sprints)